



ME 507: Micro and Nanoscale Fluid Mechanics

Course Code	ME 507
Course Name	Micro and Nanoscale Fluid Mechanics
L-T-P-C	3-0-0-3
Prerequisites	IC 142, ME 210, ME 303
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Introduction**
Overview, physics of miniaturization, scaling laws and continuum model, engineering applications of micro/fluidics. [4 Lectures]
- **Microscale fluid mechanics**
Navier Stokes equations, energy and species transport equations, constitutive relations, surface tension, Young-Laplace equation, velocity and stress boundary conditions at interfaces, exact solutions, flow regimes, inter-molecular forces, kinetic theory of gases, slip theory, Low Re flows, High Pe flows, Couette flow, Poiseuille flow, Stokes drag on a sphere, time-dependent flows, Hydraulic circuit analysis. [16 Lectures]
- **Electrokinetics**
Electrostatics and electro hydrodynamics fundamentals, electro-osmosis, electrophoresis, Dielectrophoresis, electro-capillarity and electrowetting effects. [10 Lectures]
- **Microfabrication techniques**
Micromachining of Silicon and Polymeric chips, Chemical etching and bonding, electron beam lithography, soft lithography, micromachining, casting, injection molding. [8 Lectures]
- **Introduction to Nanofluidics**
Unidirectional transport in nanochannels, transport through nanostructures with interfaces, molecular dynamics simulations, nanofluidic energy conversion [4 Lectures]

References

1. P. Tabeling, **Introduction to Micro Fluidics**, Oxford 2005.
2. B. Kirby, **Micro and Nanoscale Fluid Mechanics: Transport in Microfluidic devices**, Cambridge University Press, 2010.
3. N.T. Nguyen, S. Wereley, **Fundamentals and applications of Microfluidics**, Artech House, 2002.
4. Marc. J. Madou, **Fundamentals of Microfabrication**, 2nd Edition, CRC Press, 2002.
5. S. Colin, **Microfluidics**, John Wiley & Sons, 2009.
6. H. Bruss, **Theoretical Microfluidics**, Oxford 2008.